

Rabin-Karp Algorithm

Text: $\overset{1}{a} \overset{2}{a} \overset{3}{a} \overset{4}{a} \overset{5}{a} \overset{6}{b}$
 $=6$ $\underline{1+1+2}=4$

Pattern: $\overset{1}{a} \overset{2}{a} \overset{3}{b}$
 $=3$ $\underline{1+1+2}=4$
 $\uparrow$ $\uparrow$
 $h(p)$ hashcode

a — 1

b — 2

c — 3

d — 4

e — 5

f — 6

g — 7

h — 8

i — 9

j — 10



Rabin-Karp Algorithm

Text: $\overset{1}{a} \overset{2}{a} \overset{3}{a} \overset{4}{a} \overset{5}{a} \overset{6}{b}$
 $n=6$ $\underline{1 + 1 + 2 = 4}$

pattern: $\overset{1}{a} \overset{2}{a} \overset{3}{b}$
 $m=3$
 $\underline{1 + 1 + 2 = 4}$
 $\uparrow$ $\uparrow$
 $h(p)$ hashcode

a — 1

b — 2

c — 3

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Text: $\overset{1}{a} \overset{2}{a} \overset{3}{a} \overset{4}{a} \overset{5}{a} \overset{6}{b}$
 $n=6$ $\underline{1 + 1 + 2 = 4}$

pattern: $\overset{1}{a} \overset{2}{a} \overset{3}{b}$
 $m=3$ $\underline{1 + 1 + 2 = 4}$
 $\uparrow$
 $h(p)$ $\uparrow$
hashcode

a — 1

b — 2

c — 3

d — 4

e — 5

f — 6

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i — 9

j — 10



Rabin-Karp Algorithm

Text: ¹a ²b ³c ⁴d ⁵a ⁶b ⁷c ⁸e
n=
2+3+5
10

pattern: ¹b ²c ³e
m=3 2+3+5=10

a—1
b—2
c—3
d—4
e—5
f—6
g—7
h—8
i—9
j—10



Rabin-Karp Algorithm

Text: ^{1 2 3 4 5 6 7 8 9 10 11}
c c a c c a e d b a
= 11

1+1+5

7

Spurious Hits

m: ^{1 2 3}
d b a
4+2+1=7

$O(mn)$

$O(n-m+1)$

a — 1

b — 2

c — 3

d — 4

e — 5

f — 6

g — 7

h — 8

i — 9

j — 10



Rabin-Karp Algorithm

Text: ^{1 2 3 4 5 6 7 8 9 10 11}
c c a c c a a e d b a
= 11
 $3 \times 10^2 + 1 \times 10^1 + 3 \times 10^0$

$$\begin{aligned} 331 - 3 \times 10^2 &= \\ 331 - 300 &= 31 \\ 31 \times 10 &= 310 + 3 = 313 \end{aligned}$$

Pattern: ^{1 2 3}
d b a = 421

$$4 \times 10^2 + 2 \times 10^1 + 1 \times 10^0$$

$$400 + 20 + 1 = 421$$

a — 1

b — 2

c — 3

d — 4

e — 5

f — 6

g — 7

h — 8

i — 9

j — 10

